Introduction to parallel computing.
Report and project preparation

November 12, 2025

1 Introduction

The exam consists of a project and an oral presentation. Each student will complete the project individually. The oral presentation involves presenting (through slides) the project and answering questions related to the project AND topics covered during the lectures. Students have to be registered for the exam. **No registered students will be asked to present at the next session.** Exception only for motivated issues. Students can choose a project from the list provided in the next section. Students must send by **14 Jan 2026 at 23.59** (first winter session) or **9 Feb 2026 at 23.59** (second winter session) the report through the form at the following URL: `https://forms.gle/R511ggfPy1TUp7Qw9` The report must satisfy the following requirements.

1. Author: Student Name, Surname, ID, email
2. Maximum 4 pages (references do not count!!!)
3. References.
4. Platform and computing system description.
5. GIT and instructions for the reproducibility
6. Format: use IEEE conference template (2-column format, main text 10pt)

Disclaimer: if the report does not match one or more requirements, it will NOT be evaluated.

1.1 Report organization

Recommendations on how to organise the report. Each report can contain the following sections (on top of the mandatory sections):

Abstract

- Summary of the project, highlighting key objectives, methods, and findings (e.g., the speed-up).

Introduction of the problem and importance

- Describe the background and importance of the problem your project addresses.
- Explain the relevance of the project in the context of parallel computing.
- Outline the main objectives of your project.

State-of-the-art

- Review current research and developments related to your project.
- Discuss existing solutions and their limitations.
- Identify the gap your project aims to fill.

Contribution and Methodology

- Elaborate on the unique contributions of your project.
- Describe the methodology used in your project, including algorithms (pseudo-code), data structures, and parallelization techniques. Use figures to describe the methodology, if any.
- Discuss the challenges faced and how they were addressed.

Experiments and System Description

- Detailed description of the computing system and platform.
- Relevant specifications or configurations (e.g., libraries and programming toolchains).
- Description of the experimental setup, procedures, methodologies and metrics used in the project.
- Discussion on how experiments are designed to test the hypotheses or achieve the objectives. For example, strong scalability to check how the code scales by increasing the number of threads.

Results and Discussion

- Presentation of results.
- Analysis and **interpretation** in context. Avoid merely describing the pictures.
- Comparison with the state-of-the-art. Notice that it does not really matter if your implementation is slower than SOTA libraries. The most important part is the awarness of the performance. Use profiler and tools.

Conclusions

- Summary of key findings and contributions.
- Discussion on impact and future work

To support claims or arguments that are not directly proven or extensively discussed in the report, ensure you provide appropriate references.

- Use reputable sources such as peer-reviewed papers, conference proceedings, or technical reports. A good starting point for your search is **Google Scholar**. See next section.
- Ensure that references are cited correctly and consistently throughout your report, following the citation style specified for the course.

If you encounter any uncertainties or have specific questions while working on your project or report, do not hesitate to reach out.

- You can contact us directly by email, and we will provide clarification or guidance as needed.

Remember, a well-referenced and clearly articulated report will strengthen your work and contribute to a higher evaluation.

To search papers

To search paper

A. Major International Conferences

B. Leading Journals

Note: All listed venues are indexed in major databases (Scopus, WoS) and have high visibility in the HPC and parallel systems community.

Acronym	Full Name and Link	Publisher
SC	International Conference for High Performance Computing, Networking, Storage and Analysis	IEEE / ACM
ISC-HPC	International Supercomputing Conference – High Performance	Springer LNCS
IPDPS	IEEE International Parallel and Distributed Processing Symposium	IEEE
PPoPP	ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming	ACM SIGPLAN
HPDC	ACM International Symposium on High-Performance Parallel and Distributed Computing	ACM
ICS	ACM International Conference on Supercomputing	ACM
Euro-Par	European Conference on Parallel and Distributed Computing	Springer LNCS
HiPC	IEEE International Conference on High Performance Computing, Data, and Analytics	IEEE
Cluster	IEEE International Conference on Cluster Computing	IEEE
PACT	International Conference on Parallel Architectures and Compilation Techniques	IEEE / ACM

Table 1: Flagship conferences in Parallel Computing and High-Performance Computing.

Journal (link)	Publisher
IEEE Transactions on Parallel and Distributed Systems (TPDS)	IEEE
ACM Transactions on Parallel Computing (TOPC)	ACM
Parallel Computing (ParCo)	Elsevier
Int. Journal of High Performance Computing Applications (IJHPCA)	SAGE
Journal of Parallel and Distributed Computing (JPDC)	Elsevier
Int. Journal of Parallel Programming (IJPP)	Springer
Concurrency and Computation: Practice and Experience (CCPE)	Wiley
Future Generation Computer Systems (FGCS)	Elsevier
Cluster Computing	Springer
Supercomputing Frontiers and Innovations (JSFI)	Open Access

Table 2: Leading journals in Parallel Computing and High-Performance Computing.